$$p(x) = \begin{cases} e^{-x/\theta}/\theta, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \theta > 0$$

№3

$\vec{X}_n$ – выборка, $n=3$

оценки параметра θ: $\tilde{\theta}_1 = \bar{x}$, $\tilde{\theta}_2 = X_{(2)}$ (второй член вариационного ряда (вар. ряд – упоряд. выборка))

а) Δ на несмещённость ($\forall$ распр вер модели $M(\tilde{\theta}) = \theta$)

$M(g(\vec{x}_n)) = \theta$

$$M\xi = \int_{-\infty}^{+\infty} x\,dF(x,\theta) = \int_0^{+\infty} x e^{-x/\theta} \frac{1}{\theta} dx = -\int_0^{+\infty} x\,d e^{-\frac{x}{\theta}} = -\left[x e^{-\frac{x}{\theta}}\Big|_0^{+\infty} - \int_0^{+\infty} e^{-\frac{x}{\theta}} dx\right] = 0 + \int_0^{+\infty} e^{-\frac{x}{\theta}} dx = \frac{-1}{\theta} e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = \theta$$

$$M\xi^2 = \int_0^{+\infty} x^2 e^{-x/\theta} \frac{1}{\theta} dx = -\int_0^{+\infty} x^2\,d e^{-x/\theta} = -\left[x^2 e^{-\frac{x}{\theta}}\Big|_0^{+\infty} - \int_0^{+\infty} e^{-\frac{x}{\theta}}\,d(x^2)\right] =$$

$$= 0 + \int_0^{+\infty} 2x e^{-\frac{x}{\theta}} dx = -2\theta \cdot \int_0^{+\infty} x \cdot d(e^{-\frac{x}{\theta}}) = -2\theta\left[x e^{-\frac{x}{\theta}}\Big|_0^{+\infty} - \int_0^{+\infty} e^{-\frac{x}{\theta}} dx\right] = +2\theta \int_0^{+\infty} e^{-\frac{x}{\theta}} dx =$$

$$= -2\theta^2 e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = 2\theta^2$$

$$\Rightarrow D\xi = 2\theta^2 - \theta^2 = \theta^2$$

1) $\tilde{\theta}_1 = \frac{1}{3}\sum_{i=1}^{3} x_i$

$M\tilde{\theta}_1 = \frac{3}{3} M\xi = \theta \Rightarrow$ несмещ

$D\tilde{\theta}_1 = \frac{3}{9} D\xi = \frac{\theta^2}{3}$

2) $\tilde{\theta}_2 = X_{(2)}$

$$\xi_{(2)} \sim \sum_{i=2}^{3} C_3^i (F(t))^i (1-F(t))^{n-i} =$$

$$= C_3^2 F^2(t)(1-F(t)) + 1(F(t))^3 = 3F^2(t) - 2F^3(t) =$$

$$= 3(1-e^{-t/\theta})^2 - 2(1-e^{-t/\theta})^3$$

$$f_{X_{(2)}}(x) = \frac{dF}{dx} = 6(1-e^{-x/\theta}) \cdot \frac{1}{\theta} e^{-x/\theta} - 6(1-e^{-x/\theta})^2 \frac{1}{\theta} e^{-x/\theta} =$$

$$= \frac{6}{\theta}(1-e^{-x/\theta}) e^{-x/\theta} \cdot e^{-x/\theta}$$

$$M\tilde{\theta}_2 = \int_0^{\infty} x \cdot \frac{6}{\theta}(1-e^{-x/\theta}) e^{-2x/\theta} dx = \frac{6}{\theta}\left[\int_0^{\infty} x e^{-2x/\theta} dx - \int_0^{\infty} x e^{-3x/\theta} dx\right] =$$

$$= \frac{6}{\theta}\left[\frac{-\theta}{2}\int_0^{\infty} x\,d e^{-\frac{2x}{\theta}} + \frac{\theta}{3}\int_0^{\infty} x\,d e^{-\frac{3x}{\theta}}\right] = -3\left(x e^{-\frac{2x}{\theta}}\Big|_0^{\infty} - \int_0^{\infty} e^{-\frac{2x}{\theta}} dx\right) + 2\left(x e^{-\frac{3x}{\theta}}\Big|_0^{\infty} - \int_0^{\infty} e^{-\frac{3x}{\theta}} dx\right) = 3\int_0^{\infty} e^{-\frac{2x}{\theta}} dx - 2\int_0^{\infty} e^{-\frac{3x}{\theta}} dx = \frac{3}{2}\theta - \frac{2}{3}\theta = \frac{5}{6}\theta$$

смещ

$$M\tilde{\theta}_2^2 = \int_0^{\infty} x^2 \frac{6}{\theta}(1-e^{-x/\theta}) e^{-2x/\theta} dx = \frac{6}{\theta}\left[\int_0^{\infty} x^2 e^{-2x/\theta} dx - \int_0^{\infty} x^2 e^{-3x/\theta} dx\right] =$$

$$= \frac{6}{\theta}\left[x^2 \frac{-\theta}{2} e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} - \int_0^\infty 2x \frac{-\theta}{2} e^{-\frac{2x}{\theta}} dx\right] - \frac{6}{\theta}\left[x^2 \frac{-\theta}{3} e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} - \int_0^\infty 2x \frac{-\theta}{3} e^{-\frac{3x}{\theta}} dx\right] =$$

$$= \frac{6}{\theta}\cdot\theta \int_0^\infty x e^{-\frac{2x}{\theta}} dx - \frac{6}{\theta}\cdot\frac{2}{3}\theta \int_0^\infty x e^{-\frac{3x}{\theta}} dx = 6\left[x e^{-\frac{2x}{\theta}} \cdot \frac{-\theta}{2}\Big|_0^{+\infty} - \int_0^\infty e^{-\frac{2x}{\theta}} \frac{-\theta}{2} dx\right] -$$

$$-4\left[x e^{-\frac{3x}{\theta}} \frac{-\theta}{3}\Big|_0^{+\infty} - \int_0^\infty e^{-\frac{3x}{\theta}} \frac{-\theta}{3} dx\right] = 3\theta \int_0^\infty e^{-\frac{2x}{\theta}} dx - \frac{4}{3}\theta \int_0^\infty e^{-\frac{3x}{\theta}} dx =$$

$$= 3\theta e^{-\frac{2x}{\theta}} \cdot \frac{-\theta}{2}\Big|_0^{+\infty} - \frac{4}{3}\theta e^{-\frac{3x}{\theta}} \frac{-\theta}{3}\Big|_0^{+\infty} = \frac{3}{2}\theta^2 - \frac{4}{9}\theta^2 = \frac{19}{18}\theta^2$$

$$D\widetilde{\theta}_2 = \frac{19}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{13}{36}\theta^2$$

Исправим оценку: $\widetilde{\theta}_2' = \frac{6}{5}\widetilde{\theta}_2$ — несмещ, $M\widetilde{\theta}_2 = \theta$

$$D\widetilde{\theta}_2' = \frac{36}{25} D\widetilde{\theta}_2 = \frac{13}{25}\theta^2$$

b) Сравним эффективность оценок

$$D\widetilde{\theta}_1 < D\widetilde{\theta}_2'$$
$$\frac{1}{3}\theta^2 < \frac{13}{25}\theta^2$$
$$25 < 39$$

$\Rightarrow \widetilde{\theta}_1$ эффективнее $\widetilde{\theta}_2'$ по опр.

c) нер-во Крамера-Рао

Пусть модель является регулярной и статистика $\widetilde{g}(\vec{X}_n)$ является регулярной оценкой $g(\theta)$

Тогда $D\widetilde{g}(\vec{X}_n) \geq \frac{(g'(\theta))^2}{nI(\theta)}$ $\forall \theta \in \Theta$

1) $\widetilde{\theta}_1$ (1) $F(x,\theta)$ непр дифф по θ на Θ

$$\frac{\partial}{\partial\theta}\int_{-\infty}^{\infty} dF(x,\theta) = \frac{\partial}{\partial\theta}\int_0^{+\infty} \underset{\frac{1}{\theta}e^{-\frac{x}{\theta}}}{\rho(x,\theta)} dx = 0 = \int_0^{+\infty} \underbrace{\frac{\partial}{\partial\theta}\rho(x,\theta)}_{-\frac{1}{\theta^2}e^{-\frac{x}{\theta}}} dx = \int_{-\infty}^{\infty} d\frac{\partial F(x,\theta)}{\partial\theta}$$

$$I(\theta) = \int_{A(\theta)} \left(\frac{\partial \ln \rho(x,\theta)}{\partial\theta}\right)^2 \rho(x,\theta) dx = \int_0^\infty \left(\frac{\partial\left(-\frac{x}{\theta} - \ln\theta\right)}{\partial\theta}\right)^2 \frac{1}{\theta} e^{-\frac{x}{\theta}} dx =$$

$$= \int_0^\infty \left(\frac{x}{\theta^2} - \frac{1}{\theta}\right)^2 \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = \int_0^\infty \frac{x^2}{\theta^4} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx + \int_0^\infty \frac{-2x}{\theta^3} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx + \int_0^\infty \frac{1}{\theta^2} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx =$$

$$= \frac{1}{\theta^4} M\xi^2 - \frac{2}{\theta^3} M\xi + \frac{1}{\theta^2} = \frac{1}{\theta^4} 2\theta^2 - \frac{2}{\theta^3}\theta + \frac{1}{\theta^2} = \frac{1}{\theta^2}$$ непр т.к. $\theta > 0$ и > 0 на Θ

$\Rightarrow$ вероятностная модель — регулярн.

1.2) рег. оценка

модель регулярна

$\widetilde{\theta}_1$ несмещ

$D\widetilde{\theta}_1$ огран на $\forall$ компакте из Θ

$\Rightarrow$ дост усл регулярности оценки

Таким образом вып усл нер-ва Крамера-Рао.

$\Rightarrow D\widetilde{\theta}_1 \geq \frac{\theta^2}{3}$

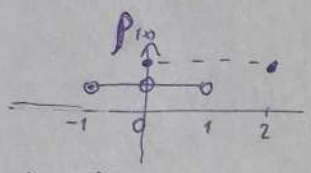
При этом $D\widetilde{\theta}_1 = \frac{\theta^2}{3}$

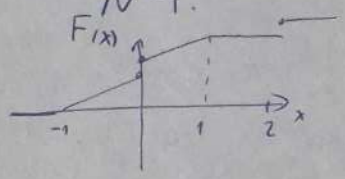
$\Rightarrow$ по ⊕ дост. усл эфф $\widetilde{\theta}_1$ — эффективная оценка

т.к. $\exists!$, то $\widetilde{\theta}_1'$ — не эфф

$\Rightarrow \widetilde{\theta}_1$ эфф чем $\widetilde{\theta}_1'$

N4.





а) $\vec{x}_n$

метод моментов:

$\xi \sim p(x) = a \times \{(-1,1)\} + b\{0;2\}$

$\int_{-\infty}^{+\infty} dF(x) = \int_{-1}^{1} a \cdot dx + 2b = 2a + 2b = 1$

$a = \theta$

$b = \frac{1}{2} - \theta$ $\qquad \theta \in (0,1)$

$\xi \sim p(x) = \theta \times \{(-1,1)\} + (\frac{1}{2} - \theta)\{0;2\}, \ \theta \in (0,1)$

вер модель

$M\xi = \int_{-\infty}^{\infty} x\, dF = \int_{-1}^{1} \theta x \cdot dx + 0 + 2(\frac{1}{2} - \theta) = 1 - 2\theta$

$M\xi^2 = \int_{-\infty}^{\infty} x^2 dF = \int_{-1}^{1} \theta x^2 dx + 0 + 2^2(\frac{1}{2} - \theta) = \frac{2}{3}\theta + 2 - 4\theta = 2 - \frac{10}{3}\theta$

$D\xi = M\xi^2 - (M\xi)^2 = 2 - \frac{10}{3}\theta - (1 - 2\theta)^2 = 1 + \frac{2}{3}\theta - 4\theta^2$

$\overline{x} = 1 - 2\theta \Rightarrow \widetilde{\theta}_1 = \frac{1 - \overline{x}}{2}$

$M\widetilde{\theta}_1 = M(\frac{1}{2} - \frac{1}{2}\overline{x}) = \frac{1}{2} - \frac{1}{2}M\xi = \frac{1}{2} - \frac{1}{2}(1 - 2\theta) = \theta$ несмещ

$D\widetilde{\theta}_1 = D(\frac{1}{2} - \frac{1}{2}\overline{x}) = \frac{1}{4}D\overline{x} = \frac{1}{4}\frac{D\xi}{n} = \frac{1}{4n}(1 + \frac{2}{3}\theta - 4\theta^2) \xrightarrow[n\to\infty]{} 0$

$\Rightarrow$ по дост усл сост.

Проверим усл Крамера-Рао:

• рег модель: $p(x, \theta)$ непр дифф на $\theta \subset \Theta$

$\frac{\partial}{\partial\theta}\int_{-\infty}^{\infty} p(x,\theta)dx = \frac{\partial}{\partial\theta} 1 = 0 \quad \int_{-\infty}^{\infty} \frac{\partial}{\partial\theta} p(x;\theta)dx = \int_{-1}^{1} \frac{\partial\theta}{\partial\theta} dx - 2\frac{\partial}{\partial\theta}(\frac{1}{2} - \theta) = 0, 0 = 0$

$I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2\right] = \int_{-1}^{1}\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2 p(x,\theta)\,dx + \sum_k \left(\frac{\partial \ln p_k(\theta)}{\partial\theta}\right)^2 p_k(\theta)$

$$I(\theta) = \int_{-1}^{1} \frac{1}{\theta^2}\cdot\theta\cdot dx + \left(\frac{1}{(1/2-\theta)}(-1)\right)^2 (1/2-\theta)\cdot 2 = \frac{1}{\theta}\cdot 2 + \frac{2}{(1/2-\theta)} =$$

$$= \frac{2}{\theta} + \frac{4}{(1-2\theta)} = \frac{2}{\theta(1-2\theta)}$$

$I(\theta) = \frac{2}{\theta(1-2\theta)}$ непр на $\theta\in(0;\frac{1}{2})$

$\Rightarrow$ модель регулярна

• рег оценки:

рег. модели

несмещ оценки

$\textcircled{D}\ \tilde{\theta}_1 = \frac{1}{4n}(1+\frac{2}{3}\theta-4\theta^2)$ огран на $\forall$ компакте из $(0,\frac{1}{2})$ $\Big\} \Rightarrow$

$\Rightarrow$ оценка регулярна по дост усл.

Дост. усл эфф:

$$\frac{1}{4n}(1+\frac{2}{3}\theta-4\theta^2) \geq \frac{1}{nI(\theta)} = \frac{1}{n}\frac{1}{\frac{2}{\theta(1-2\theta)}} = \frac{\theta(1-2\theta)}{2n}$$

$$\frac{1}{4n} + \frac{1}{6n}\theta - \frac{1}{n}\theta^2 \geq \frac{1}{2n}\theta - \frac{1}{n}\theta^2$$

$\Rightarrow$ не выполняется достат. усл эффективности

Метод максимального правдоподобия

$\xi \sim p(x) = \theta\, x\{-1,1\} + (\frac{1}{2}-\theta)\{0;2\}$, $\theta\in(0,1)$

$\vec{x_n}$ выборка m чисел встретилось 0 или 2

$n-m$ из $\{-1,1\}\setminus\{0\}$

ф-ция правдоподобия:

$$L(\theta) = \prod_{i=1}^{n} p(x_i,\theta) = \theta^{n-m}(\frac{1}{2}-\theta)^m$$

$$\frac{\partial}{\partial\theta}(\ln L) = \frac{\partial}{\partial\theta}\left((n-m)\ln\theta + m\ln(\frac{1}{2}-\theta)\right) = \frac{n-m}{\theta} - \frac{m}{\frac{1}{2}-\theta}$$

$$\frac{n-m}{\tilde{\theta}} - \frac{m}{\frac{1}{2}-\tilde{\theta}} = 0$$

$$\frac{(n-m)(\tilde{\theta}_2-\frac{1}{2})+m\tilde{\theta}_2}{\tilde{\theta}_2^2-\frac{1}{2}\tilde{\theta}} = \frac{n\tilde{\theta}_2-\frac{1}{2}(n-m)}{\tilde{\theta}_2^2-\frac{1}{2}\tilde{\theta}_2} = 0 \Rightarrow \tilde{\theta}_2 = \frac{1}{2}(1-\frac{m}{n}) = \frac{1}{2}(1-\nu)$$

пров на max: $(\ln L)''_{\theta\theta} = \frac{-(n-m)}{\theta^2} - \frac{m}{(\frac{1}{2}-\theta)^2} = -\left(\frac{m(4\theta-1)+n(2\theta+1)^2}{\theta^2(1-2\theta)^2}\right) < 0$

$\Rightarrow (\ln L)''_{\theta\theta} < 0 \Rightarrow$ max

$\tilde{\theta}_2 = \frac{1}{2}(1-\nu)$ ν частота появл (0 или 2)

$M\tilde{\theta}_2 = \frac{1}{2} - \frac{1}{2}M\nu = \frac{1}{2} - \frac{1}{2}p = \theta$ несмещ оценка

$D\tilde{\theta}_2 = \frac{1}{4}D\nu = \frac{1}{4}\frac{p(1-p)}{n} = \frac{1}{4}\frac{(1-2\theta)2\theta}{n} = \frac{\theta(\frac{1}{2}-\theta)}{n} \xrightarrow[n\to\infty]{} 0$

явл сост по дост усл.

модель рег.
оценка несмещ
$D\tilde{\theta}_2$ огран на $\forall$ компакте из $(0,\frac{1}{2})$

$\Rightarrow$ оценка регулярна по дост усл

<u>Сравним оценки на эффективность с помощью нер-ва Крамера-Рао</u>

$D\tilde{\theta} \geq \frac{1}{nI(\theta)} = \frac{1}{n\frac{2}{\theta(1-2\theta)}} = \frac{\theta(\frac{1}{2}-\theta)}{n}$

$D\tilde{\theta}_2 = \frac{\theta(\frac{1}{2}-\theta)}{n}$

$\Rightarrow \tilde{\theta}_2$ — эффективная оценка по дост усл

т.к. если эфф оценка $\exists$, то она !, то $\tilde{\theta}_1$ не явл эфф